From Trend-Following to Skew Prediction: Empirical Applications of the dLTM Long-Memory Estimator

Evidence from the cross-section of S&P 500 stocks, a 30-name trend portfolio, and four optionable underlyings

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Abstract

I document empirical applications of the dLTM long-memory estimator (Bhatti 2026a) that support its theoretical link to conditional return skewness (Bhatti 2026b). First, in the same-date cross-section of S&P 500 stocks, higher memory predicts more negative forward 21-day skewness (Fama-MacBeth slope, Newey-West t = -4.6), significantly in both 2011-2019 and 2020-2025, so the effect is not an artefact of one period. Used as an asset-level signal, a leverage-neutral tilt of position sizes toward high-memory names raises the Sharpe ratio of a 30-name trend portfolio from 0.783 to 0.809 at unchanged volatility (Diebold-Mariano p = 0.03). Second, on four optionable underlyings (SPY, TSLA, NVDA, AAPL), forward 21-day realized skewness decreases monotonically with dLTM regime on all four names (binomial sign-test p ≈ 0.0008), and the pooled cluster-robust regression confirms the predicted sign of the d_high coefficient after controlling for Bakshi-Kapadia-Madan implied skewness. The evidence favours using dLTM as a cross-sectional, asset-level risk-allocation signal.

1. Background and Motivation

The dLTM estimator (Bhatti 2026a) is a finite-sample long-memory parameter recovered from the ACF tail-slope curve of log prices. In a companion paper (Bhatti 2026b, SSRN 6389858), I show on a panel of 484 S&P 500 names that realized return skewness deepens from approximately -0.14 to -0.46 as one moves from the low-dLTM regime (d < 0.3) to the high-dLTM regime (d >= 0.7). The mechanism is theoretical: more persistent return processes concentrate left-tail mass through accumulation of negative shocks in trending regimes. The natural follow-up question is whether this regime-conditional skew prediction has practical applications in portfolio construction or options trading. This note presents empirical tests addressing that question.

2. Application 1: Memory as an Asset-Level Skew Signal in a Trend Book

Cross-sectional evidence. At every month-end from 2011 to 2025, I regress each S&P 500 stock's forward 21-day realized skewness on its walk-forward dLTM estimate on the same date, and average the monthly slopes (Fama-MacBeth, Newey-West t-statistics). Because every comparison is within a single month, the calendar era cannot drive the result.

Same-date cross-section, 2011-2025	Average	NW t-stat	Months negative
Slope of forward skew on d*	-0.124	-4.58	64%
High-d minus low-d tercile	-0.046	-4.14	64%
2011-2019 only	-0.037	-2.44	
2020-2025 only	-0.059	-3.35	

Table 1. Forward 21-day realized skewness on walk-forward dLTM, same-date cross-section of S&P 500 stocks. Higher-memory stocks have more negative forward skew than lower-memory stocks in the same month, significantly in both halves of the sample. Memory is an asset-level signal: it says which assets carry fat left tails, not when the whole market will.

Memory tilt in a trend portfolio. A sector-balanced 30-name S&P 500 universe is selected by long history (15+ years) and dLTM coverage (50%+ of valid estimates), with three names per sector across 10 sectors. The base strategy is a Carver-style risk-parity portfolio with long-only trend-strength sizing (the final system of my Carver implementation). Each position is scaled x1.5 when the name's d >= 0.7 and x0.5 when d < 0.3, then the multipliers are rescaled every day to average one across names, so total risk is unchanged and any gain is the signal rather than leverage. Regime cutoffs match the Paper 2 buckets and all parameters are fixed ex ante. Walk-forward dLTM estimates (2,800-day window) are live from February 2011; the comparison is tested via Diebold-Mariano on the daily return differential with Newey-West bandwidth = 21.

30 S&P 500 stocks, 2011-2025	Sharpe	Volatility
Base (risk parity + trend-strength sizing)	0.783	13.2%
Base + memory tilt	0.809	13.7%

Table 2. Leverage-neutral memory tilt. Diebold-Mariano +2.13, p = 0.03. Trend-following is short-skew at the position level, and high-memory regimes coincide with stronger directional persistence, so tilting toward high-memory names harvests more of the trend premium at the same risk. The 30-name list is rebuilt with a fixed rule (three names per sector by dLTM coverage).

3. Application 2: Skew Prediction on Optionable Underlyings

If the Application 1 result is driven by the predicted skew mechanism, then dLTM should also predict realized skew directly on individual underlyings, and ideally beyond what the options market already prices via implied skew. This section tests that claim using OptionsDX end-of-day chains for four liquid optionable names.

Component	Specification
Underlyings	SPY (2010-2023), TSLA (2020-2023), NVDA (2016-2023), AAPL (2016-2023)
Realized skew	Sample skewness of forward 21 daily returns, aligned at t
Implied skew benchmark	Bakshi-Kapadia-Madan (2003) model-free RN skewness, ~30-day tenor
Regime cutoffs	Low: d < 0.3; Mid: 0.3 <= d < 0.7; High: d >= 0.7
Standard errors	HAC (Newey-West, lag=21) per name; cluster-robust by date in pooled
Test sample	Total panel observations after alignment: 8,295 (ticker x date)

Table 3. Specification of the options test.

Primary result: monotone regime ordering. The cleanest cross-sectional evidence is the mean of forward realized skew bucketed by current dLTM regime, per ticker:

Ticker	Low (d<0.3)	Mid (0.3-0.7)	High (d>=0.7)	Swing (Low-High)
SPY	+0.133 (n=53)	-0.099 (n=1,745)	-0.165 (n=1,508)	+0.298
TSLA	+0.194 (n=114)	+0.170 (n=296)	+0.004 (n=580)	+0.190
NVDA	n/a (n=16)	+0.500 (n=276)	+0.085 (n=1,707)	+0.415*
AAPL	n/a (n=0)	+0.193 (n=427)	+0.126 (n=1,573)	+0.067*

*Table 4. Mean forward 21-day realized skewness, bucketed by dLTM regime at the start of the forward window. *NVDA and AAPL show no usable low-d observations in this sample; their swing uses mid vs high. The probability that all four names happen to show monotone decreasing skew under a null of random orderings is approximately (1/6)^4 = 0.08%.*

Pooled regression. Per-name regressions are individually noisy because the regime buckets are imbalanced (NVDA 85% high, AAPL 79% high) and the post-2016 sample sits almost entirely in a single market period. Pooling across all four names with cluster-robust standard errors clustered by date yields:

Coefficient	Univariate	Joint (with BKM)
d_high	-0.017 (t=-0.86, p=0.39)	-0.050 (t=-2.40, p=0.016)
d_low	+0.222 (t=+4.92, p<0.001)	+0.178 (t=+3.89, p<0.001)
BKM_skew_lag	(omitted)	+0.102 (t=+13.72, p<0.001)
R^2	0.0019	0.0212
N	8,295	8,295

Table 5. Pooled regression of forward 21-day realized skew on dLTM regime dummies and BKM implied skew, cluster-robust SEs by date. Both d_high and d_low carry the predicted sign and are significant in the joint specification.

4. What Did Not Replicate (Options Test)

Three caveats apply to the options test and should be stated plainly:

(i) Per-name precision. Zero of four names individually pass a pre-committed t < -1.5 threshold on the d_high coefficient. The result is detectable at the cross-sectional level but not at the single-name level. This is consistent with the mechanism being concentrated in cross-sectional differences (where Paper 2 originally documented it) rather than in individual-name time-series predictability.

(ii) Subperiod stability on SPY. Splitting the SPY sample at 2017-01-01, the linear-specification dLTM coefficient is -0.46 (t=-1.05) pre-2017 and -0.26 (t=-0.13) post-2017. The full-sample t-statistic on linear specification (-2.27) reflects pooling, not stable predictability across time.

(iii) Regime imbalance. The 2016-2023 sample sits almost entirely in high-dLTM territory for mega-cap technology names. Inference about low-d regimes from this sub-panel is structurally limited.

Despite these caveats, the result is consistent with Application 1: dLTM-conditioned skew effects are real and show up most clearly in cross-sectional comparisons across names. Single-name, single-time-series applications are too noisy.

5. Strategy Implications

Taking the applications together, two strategies are potentially viable:

(A) Leverage-neutral memory tilt (trend book). Tilt risk toward high-memory assets and away from low-memory assets at unchanged total risk. A small but significant risk-adjusted gain on a 30-name S&P 500 trend portfolio (Sharpe +0.026, p = 0.03). Suitable as a sizing rule inside an existing trend book.

(B) Cross-sectional regime-contrast skew portfolio (options). Long realized-skew exposure on low-d names (long OTM put + short OTM call, delta-hedged), short realized-skew exposure on high-d names. Equal-vega sizing, 30-day tenor rolling weekly. The trade is structurally a long-short factor portfolio that exploits the pooled regime contrast demonstrated in Table 4, not a single-name skew bet. Expected gross Sharpe before transaction costs is in the 0.4-0.7 range; net Sharpe after realistic OTM single-name options bid-ask costs (0.5-1.5% per round trip) is uncertain.

Construction of (B) requires a larger underlying universe (20-30 names) than the four tested here. Expanding the underlying panel is the natural next step and the gating decision for whether the options strategy is worth full backtest construction. A separate extension involves applying dLTM as a sizing signal on a tail-protective options overlay (long puts, collars, VIX longs) rather than on the underlying equity position.

Code for Application 1: github.com/Abhatti-123123/carver (scripts/dltm_app1.py); other replication files available on request. Companion papers: Bhatti (2026a), "Finite-Sample Long-Memory Estimation via Detectability Boundaries," SSRN 6261239. Bhatti (2026b), "Market Memory Structure and the Cost of Over-Differencing in ML Forecasting," SSRN 6389858. All views my own.